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Hydrogen and electricity production from Mount Cameroon's energy potential: Seismic imaging, techno-environmental-socio-economic assessment, and sensitivity considerations of a zero-carbon Fuel cell/Geothermal/Electrolyzer hybrid system

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ABSTRACT

Geothermal energy represents a sustainable and low-carbon resource with strong potential to support decentralized energy systems in volcanic regions. In underexplored areas such as Mount Cameroon, accurate assessment and optimal utilization of geothermal resources are critical for reliable energy supply. Recent global seismic tomography models indicate a pronounced shear-wave low-velocity anomaly beneath Mount Cameroon, suggesting elevated subsurface temperatures consistent with geothermal potential. Building on this geophysical evidence, this study proposes an off-grid hybrid energy system integrating geothermal power, and fuel cell technology to simultaneously supply electricity and hydrogen. A comprehensive techno-economic, environmental, and sensitivity considerations were taken into account using HOMER Pro to identify the optimal system configuration under local demand conditions. The load profile shows recurrent peaks, requiring a flexible and dispatchable hybrid system. Results indicate that the optimal configuration comprises a 110 kW geothermal steam turbine, a 50 kW fuel cell, a 10 kW electrolyzer, a 25.5 kW power converter, and a 20 kg hydrogen storage tank. This system achieves reliable power supply with a net present cost of approximately US$ 1.84 million and a levelized cost of energy of US$ 0.465/kWh, while ensuring zero direct carbon emissions. Sensitivity analyses reveal that system economics are most influenced by fuel cell capital and replacement costs. The proposed hybrid system demonstrates the technical feasibility and economic viability of coupling geothermal energy with hydrogen technologies in volcanic regions. The results highlight its potential as a replicable solution for clean, resilient, and decentralized energy access in remote and underserved communities.

1. Introduction

The growing demand for sustainable and resilient energy solutions has driven global interest in harnessing renewable resources for electricity and hydrogen production [1]. Geothermal energy, particularly in volcanic regions, offers a stable and abundant heat source that can be efficiently utilized for power generation and hydrogen production when integrated into hybrid energy systems [2]. Mount Cameroon, a volcanic hotspot, presents significant untapped geothermal potential, yet its energy landscape remains dominated by fossil fuels and unreliable grid access [3]. In parallel, hydrogen has emerged as a promising energy carrier, providing long-term storage solutions and enabling a decarbonized energy future [4]. However, effective deployment of hydrogen-based systems requires a reliable and continuous power supply, which can be achieved through innovative hybrid configurations [5]. Different architectures of hybrid systems including geothermal energy were developed.

Lentz and Almanza designed a hybrid system combining solar and geothermal energy to improve steam production within the geothermal cycle by integrating a parabolic trough solar field [6]. The study revealed that this setup enhanced the system's capacity factor by supplying extra steam during periods of high energy demand [6].

Ghasemi et al. designed a performance-based model for an organic Rankine cycle operating with low-temperature geothermal brine [7]. The model was tested against a dataset comprising 7200 operational records collected over a year. Optimization was achieved by adjusting the number of active turbines to enhance efficiency. A solar trough

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Nomenclature

P_{FC}	Power generated from the fuel cell [W]
η_{FC}	Efficiency of the fuel cell
$P_{HT/FC}$	Power delivered from the hydrogen storage unit to the fuel cell system [W]
P_E	Power conveyed from the electrolyzer system to the hydrogen storage unit [W]
η_E	Efficiency of the elecytrolyzer
$P_{REN/E}$	Power delivered to the electrolyzer originating from clean energy inputs [W]
HT_E	Amount of energy accumulated within the hydrogen storage unit [J]
η_{HT}	Efficiency of the hydrogen tank
Δt	Time step during the computation [s]
P_{conv}	Power of the converter [W]
$P_{m,l}$	Peak electrical demand [W]
η_{inv}	Efficiency of the inverter
P_{out}	Electrical output power of a steam turbine generator [W]
$\dot{m}$	Mass flow rate of steam [kg/s]
h_{in}	Specific enthalpy of the steam at the turbine inlet [kJ/kg]
h_{out}	Specific enthalpy of the steam at the turbine outlet [kJ/kg]
η_t	Turbine efficiency (typically between 0.7 and 0.9)
NPC	Net present cost [US$]
COE	Cost of energy [US$/kWh]
i	Annual real discount rate [%]
R_{proj}	Project lifetime [years]
E_s	Total electrical load delivered [kWh/year]
$C_{an.t}$	Total yearly cost of the entire system [US$/year]
CRF	Factor used to spread capital cost over the system's operational lifespan [%]

system operating at low temperatures was also integrated to improve the system's overall performance. The hybrid approach resulted in a notable increase in second-law efficiency, achieving up to 3.4% higher efficiency than the individual geothermal and solar setups.

A study was conducted to examine the potential of hybrid solar-geothermal power plants in increasing electricity generation while reducing the influence of daily temperature fluctuations [8]. Using Aspen-HYSYS, various plant configurations were modeled and analyzed thermodynamically for a reservoir temperature of 120 °C and a steady brine flow rate of 50 kg/s. Under Australian climate conditions, where summer temperatures reach approximately 31 °C, findings showed that a hybrid system would generate more power than independent solar or geothermal plants, provided that at least 68% of the total energy input came from solar energy.

Liu et al. investigated a power generation system that integrated geothermal energy with fossil fuel sources, combining a supercritical 1000 MW power unit with a geothermal feedwater preheating mechanism [9]. Their study revealed that when geothermal resource temperatures ranged from 140 to 160 °C, the hybrid geothermal-coal configuration produced significantly more power than an optimized organic Rankine cycle system utilizing isopentane, with an increase of approximately 90% at a geothermal source distance of 0 km and 39–49% at 20 km.

A comprehensive model was designed to assess the performance of a hybrid system combining solar energy with a binary geothermal cycle [10]. The configuration integrated an organic Rankine cycle powered by low-temperature geothermal brine alongside a solar trough field. The findings demonstrated that incorporating solar energy into the system lowered the levelized cost of electricity by 2% compared to a purely geothermal-based setup.

A renewable energy system was designed by integrating wind, solar, and geothermal sources to enhance electricity generation from low-temperature geothermal resources [11]. To maximize energy utilization, a storage system was incorporated to facilitate hydrogen production from surplus electricity. The system successfully supplied an annual electricity demand of 1983.040 MWh while generating an excess of 362.077 MWh, leading to a total production of 2402.572 MWh. The surplus energy was efficiently converted into hydrogen, yielding an annual output of 3423 kg. The analysis determined the levelized cost of energy for the hybrid configuration to be US$ 0.561/kWh.

A dual-section hybrid energy system was developed, optimized, and implemented to enhance the sustainability of wind power [12]. The power generation unit included a wind turbine, an alkaline electrolysis system, a biodiesel generator, and a proton exchange membrane fuel cell. Meanwhile, the water supply unit for the electrolysis and fuel cell operations incorporated a ground-source heat pump, a microalgal culture pond, a parabolic trough collector, and a solar still desalination system. The findings indicated that this configuration significantly improved the wind turbine capacity factor by 82.3%, with a levelized cost of electricity of $0.161 per kWh.

A technical evaluation was carried out on a system integrating a ground-source heat pump with photovoltaic thermal collectors and a phase-change material storage tank to meet the energy needs of a residential building [13]. The findings revealed that the irreversibility of the solar modules in the heat pump-photovoltaic thermal collector-phase-change material configuration was 6.6% lower compared to the heat pump-photovoltaic setup. Additionally, the annual energy demand analysis indicated that incorporating collectors into the hybrid system reduced the total load of the building by 6.5%.

Clise et al. presented a dynamic simulation model of an innovative solar-geothermal polygeneration system, along with its exergetic and exergoeconomic assessments [14]. The plant was designed to provide electricity, heating, cooling, and freshwater to a small community connected to a district energy network. The hybrid system integrated an Organic Rankine Cycle powered by medium-enthalpy geothermal energy and a Parabolic Trough Collector solar field. The results indicated that the overall exergy efficiency ranged between 40% and 50% in "Thermal Recovery Mode" and between 16% and 20% in "Cooling Mode."

Liu et al. conducted a comprehensive evaluation of a nanofluid-enhanced geothermal-photovoltaic hybrid system, addressing challenges related to multi-objective optimization and multi-criteria assessment [15]. Their findings indicated that the optimal configuration included a 2% Al_2O_3 nanofluid combined with a spiral pipe. The integration of monocrystalline solar panels with a nanofluid-assisted heat pump resulted in the lowest carbon dioxide emissions (550 kg/year), the most cost-effective energy production (198.18 US$), and the highest exergy efficiency. However, the hydrogen production cost remained relatively high at 211.78 US$/MWh.

A hybrid energy system was developed, integrating five photovoltaic thermal solar panels, a 300-meter geothermal loop, and 9463.53 kg of phase change material thermal battery storage to meet the energy needs of a 325 m^2 residential building in Oshawa, Canada [16]. The maximum heating and cooling loads of the building were 13.8 kW and 8.7 kW, respectively. A thermodynamic analysis was conducted for January and the entire year. The results indicated that in January, the solar panels supplied 8 W of thermal energy and 50 W of electrical power, while geothermal and thermal storage contributed 16.8 kW and 9 kW over the year, respectively. Additionally, the system required an extra heating load of 1.85 kW from the furnace. The overall energetic and exergetic coefficients of performance were determined to be 54.58% and 3.34% in winter and 42.6% and 4.47% in summer, respectively.

A hybrid system integrating solar and geothermal resources for hydrogen production, power generation, heating, and cooling was developed and evaluated for practical applications [17]. The proposed renewable energy system included solar photovoltaic thermal modules for heating, water heating, and hydrogen production, while geothermal energy was utilized for electricity generation, cooling, and hydrogen production. Energy and exergy analyses were performed to assess the system's performance, and the influence of various parameters on the energy and exergy efficiencies of both the overall system and its subsystems was examined. The results indicated that the system achieved maximum energy and exergy efficiencies of 10.8% and 46.3%, respectively, at a geothermal water temperature of 210 °C. Additionally, the impact of varying geothermal water temperature and employing different working fluids on system performance was investigated.

Noorollahi et al. investigated the integration of geothermal energy with solar power [18]. The simulation focused on extracting geothermal heat from an abandoned oil well in the Ahwaz oil field, alongside the design of a binary cycle power plant. The combination of concentrated solar thermal energy with geothermal sources led to a 23.5% increase in annual power generation.

A hybrid biomass-geothermal system designed to provide heat in low-enthalpy regions with extremely cold climates was optimized using a nonlinear optimization approach [19]. The findings strongly demonstrated the potential for more cost-effective operation of hybrid energy systems by incorporating storage solutions to integrate with smart thermal grids or district heating networks.

A grid-connected hybrid system integrating geothermal, photovoltaic, and wind energy was evaluated for Tattapani, Azad Jammu Kashmir, Pakistan, where geothermal energy from hot springs was sufficient to meet the community's continuous base load demand [20]. The analysis indicated that a feasible system configuration consisted of a 250 kW geothermal unit, a 250 kW solar PV array, and a 100 kW wind turbine. The proposed design resulted in a net present cost of 234.11 million rupees, assuming an interest rate of 5%.

Abbasi et al. analyzed the efficiency of a ground source heat pump integrated with a photovoltaic system to supply the heating and cooling needs of a zero-energy residential building [21]. The findings revealed that implementing this hybrid system led to an annual reduction of approximately 70 tons of carbon dioxide emissions.

Naseh and Behdani developed an optimization approach for determining the appropriate sizing of a geothermal/PV/wind/diesel system in both off-grid and grid-connected setups [22]. The study indicated that inadequate management of geothermal reservoirs could lead to their depletion over time.

Corigliano et al. investigated the use of the organic Rankine cycle for converting waste heat from data centers into electricity [39]. Based on an extensive review of existing studies, the work analyzed energy recovery strategies with particular attention to power generation potential, working fluid selection, and system design criteria. The authors first assessed both the quantity and quality of recoverable waste heat, noting that air-based cooling systems are the most widely deployed due to their simplicity, representing the majority of current installations, while water-based cooling offers higher thermal efficiency but is less frequently implemented. A detailed evaluation of candidate organic working fluids was carried out, accounting for the thermal constraints imposed by the available heat source for evaporation and the cooling medium for condensation. The study then examined the configuration of an organic Rankine cycle system, including internal fluid circuits, and proposed a conceptual case study simulated using the DWSIM software environment. In this configuration, warm air or hot return water from the data center cooling loop was used as the heat source for fluid evaporation, while ambient water was employed for condensation. Pentane and isopentane were identified as suitable working fluids, with pentane-based systems achieving net electrical efficiencies between approximately three and seven percent as pressure ratios increased, and isopentane-based systems exhibiting comparable performance over a similar operating range. In addition, the study quantified key performance indicators for a reference data center, including power usage effectiveness, energy reuse factor, energy reuse effectiveness, and reductions in greenhouse gas emissions. The work concluded by providing practical guidelines for system assessment, incorporating exergy analysis and outlining procedures for sizing heat exchangers such as evaporators and condensers.

De Lorenzo et al. analyzed the thermal behavior of a solid oxide electrolyser by developing a high-accuracy numerical model capable of describing both transient and steady-state operating conditions [40]. The approach explicitly accounted for thermal energy supplied by a concentrated solar source, incorporating solar radiation data to support the sizing and control of the heat input required to maintain stable operation. By coupling the thermal model with a thermo-electrochemical description of the electrolyser, the study evaluated temperature distributions, heat requirements, losses to the environment, and the potential for internal heat recovery. Particular attention was given to start-up and shut-down phases, where strict temperature and thermal-gradient constraints must be respected to ensure safe operation. The methodology was validated through application to an experimental solid oxide electrolyser system at the University of Calabria, and the results showed that careful, adaptive thermal management is essential during transient phases to comply with operational limits and preserve system integrity.

Corigliano and Fragiacomo presented a numerical modeling framework for the energy and environmental assessment of a hydrogen-based power station integrating proton exchange membrane electrolysis and fuel cell units with a hybrid hydrogen storage system combining compressed gas and metal hydrides [41]. The study proposed a structured methodology for system design and simulation, emphasizing the interaction between the hydrogen power unit and an external electrical network under coordinated operating strategies. Thermal management and auxiliary systems were explicitly included to capture realistic operating behavior. Both steady-state and transient simulations were carried out in a dynamic simulation environment to evaluate system performance under different energy supply scenarios. The analysis quantified hydrogen flows, energy balances, storage state of charge, temperature evolution, and carbon dioxide emissions. By comparing fossil-based, mixed, and fully renewable electricity supply cases, the results demonstrated substantial reductions in carbon emissions when renewable power is used, alongside electrolyzer and fuel cell efficiencies consistent with practical operation, thereby highlighting the environmental benefits of renewable-powered hydrogen energy systems.

Corigliano et al. reported that PEM fuel cells and PEM electrolyzers had gained momentum in renewable microgrids, alongside intensified work on safe, efficient hydrogen storage [42]. They modeled and simulated a 1 kW grid-connected H2PEM station with 5 Nm^3 dual storage (compressed H_2 + metal hydrides), implemented a control strategy in Matlab/Simulink, and showed fast dynamics, accurate power tracking, effective H_2 charge/discharge management, and resilient operation under varied scenarios.

Despite significant advancements in hybrid geothermal energy systems, most existing studies have primarily focused on integrating geothermal with solar, wind, or biomass sources, often overlooking the potential of hydrogen as an energy carrier. Additionally, while research has demonstrated the feasibility of geothermal-based power generation, few studies have explored its deployment in volcanic regions such as Mount Cameroon, where seismic anomalies indicate substantial geothermal potential. This study introduces an innovative fuel cell/geothermal/electrolyzer hybrid system tailored to harness the region's geothermal heat for simultaneous electricity and hydrogen production. Unlike previous works, this research leverages seismic imaging to assess the geothermal potential and integrates hydrogen storage, ensuring system flexibility and energy security. Using HOMER Pro, a comprehensive techno-environmental-socio-economic and sensitivity considerations are taken into account to determine the system's optimal configuration and viability. The findings highlight the feasibility of a zero-carbon, decentralized energy solution capable of addressing energy access challenges in the region while supporting hydrogen infrastructure development for a sustainable future.

This paper is structured as follows: Section 2 presents the mathematical modeling of the hybrid system components, economic modeling, and the case study, which includes the study area description, geothermal potential assessment through seismic imaging, environmental and social evaluation, and sensitivity considerations. Section 3 discusses the results, covering seismic imaging findings, the optimal hybrid system configuration, electricity and hydrogen production, environmental and social impacts, and sensitivity analysis. Finally, Section 4 provides the conclusion and outlines future perspectives.

2. Methods

This section outlines the methodological framework used to design, model, and evaluate the hybrid energy system, integrating technical, economic, environmental, and social dimensions.

2.1. Mathematical modeling

This section presents the mathematical models used to simulate the performance and behavior of the main components of the hybrid system. Each subsection describes the relevant equations, assumptions, and parameters associated with the Fuel Cell, Electrolyzer, Hydrogen Tank, Converter, and Steam Turbine Generator. Additionally, a dedicated part is provided for the economic modeling approach used to assess system costs and financial viability.

The objective of this work was system-level design and annual techno-economic optimization of a hybrid geothermal–hydrogen architecture using HOMER Pro. For this purpose, the component models were intentionally formulated at an energy-balance (quasi-steady) level, linking electrical power, conversion efficiencies, hydrogen production/consumption, and storage state evolution. This level of detail captures the dominant interactions that govern sizing and dispatch decisions while keeping the computational burden tractable for the large number of simulations required by the optimization and sensitivity analyses. Detailed electrochemical, thermal, start-up/shut-down transients, and degradation sub-models were therefore not included because they require proprietary or unavailable parameters for the selected commercial units and would exceed the scope of a planning-oriented assessment. The main assumptions and limitations of this simplification are stated explicitly in each component subsection and discussed as a limitation of the study.

2.1.1. Fuel cell

Electricity production in fuel cells is achieved through an electrochemical mechanism that transforms hydrogen's stored energy into usable electrical output without relying on combustion. This mechanism involves a reaction where hydrogen molecules are oxidized at one electrode, releasing electrons that flow through an external path to create electric current, while oxygen is reduced at the opposite electrode, yielding water as a secondary product. In contrast to traditional power systems, this method offers greater efficiency, minimal environmental impact, and uninterrupted performance as long as reactants are continuously supplied. Functionally, fuel cells resemble batteries but differ in their use of a constant fuel stream instead of periodic charging. Several variables, including hydrogen quality, system temperature, and membrane performance, determine the overall efficiency and capacity of the device. When integrated into renewable-based hybrid systems, fuel cells significantly improve stability and sustainability of energy supply. The electric output depends on factors such as hydrogen availability, the effectiveness of the conversion process, and operational parameters. This study uses a polymer electrolyte membrane configuration, and the electrical generation is represented mathematically by Eq. (1a) [23]:

$$P_{FC} = \eta_{FC} \times P_{HT/FC} \tag{1a}$$

where P_{FC} is the fuel-cell electrical power output (W), η_{FC} is the fuel-cell efficiency, and $P_{HT/FC}$ is the power equivalent of the hydrogen energy flow supplied to the fuel cell (W). In the simulation, $P_{HT/FC}$ was bounded by the rated capacity and dispatched only when stored hydrogen was available. It is defined in Eq. (1b):

$$P_{HT/FC} = \dot{m}_{H_2,FC} \bullet LHV_{H_2} \tag{1b}$$

where $\dot{m}_{H_2,FC}$ is the hydrogen mass flow to the fuel cell (kg/s) and LHV_{H_2} is the lower heating value of hydrogen (kWh/kg or MJ/kg).

In HOMER Pro, the fuel cell was modeled as a DC generator, and its hydrogen consumption was specified through a fuel-consumption curve; HOMER then derives the corresponding efficiency curve as a function of load. In this planning-level model, η_{FC} was treated as a nominal electrical efficiency (LHV basis), consistent with typical PEMFC values reported in the literature (commonly in the 40–60% range), and implemented consistently via the hydrogen fuel curve parameters used by HOMER.

A fuel cell with a capacity of 50 kW was implemented in the system [24]. Its technical characteristics are presented in Table 1.

2.1.2. Electrolyzer

An electrolyzer functions by breaking down water molecules into hydrogen and oxygen using electrical energy. This electrochemical operation involves applying a current to water, generating hydrogen gas at the negative electrode and oxygen at the positive electrode. It plays a key role in hydrogen generation, particularly in renewable-based systems where surplus electricity from sources like geothermal or solar energy can be stored as hydrogen. Unlike a fuel cell, which uses hydrogen to produce electricity, the electrolyzer requires electric input to produce hydrogen. Its effectiveness is influenced by factors such as electrode composition, conductivity of the membrane, system temperature, and the quality of the supplied current. The volume of hydrogen generated depends on the input power, conversion efficiency, and electrochemical properties, generally modeled using current, voltage, and Faraday's principles. The energy directed from the electrolyzer to the hydrogen storage unit is calculated using equation (2a) [23].

$$P_E = \eta_E \times P_{REN/E} \tag{2a}$$

where P_E is the power directed to hydrogen production/storage (W), η_E is the electrolyzer efficiency, and $P_{REN/E}$ is the electrical power supplied to the electrolyzer from the generation system (W). The electrolyzer operation was limited by its rated capacity and activated during surplus generation periods.

$P_{REN/E}$ can be written in Eq. (2b):

$$P_{REN/E} = \dot{m}_{H_2,El} \bullet LHV_{H_2} \tag{2b}$$

Table 1
Technical characteristics of the fuel cell [24].

Parameters	Values
Electrolyte	PEMFC
Fuel type	Hydrogen Oxygen Fuel Cell
Hydrogen Purity	99.99%
Voltage range	27–300 V
Working Temperature	<100 °C
Rating power	50 W–150 kW
Rating current	4.2–650 A
Cooling	Liquid cooled
Working environment temperature	−30 °C-50 °C
Environment humidity	0–95%
Capital cost	40,000 US$
Replacement cost	40,000 US$
Operation and maintenance cost	100 US$

where $\dot{m}_{H_2,El}$ is the hydrogen mass flow rate generated by the electrolyzer (kg/s).

In HOMER Pro, the electrolyzer was represented using its rated capacity and performance parameters (efficiency and/or hydrogen production curve/specific energy consumption), from which the hydrogen production rate is computed as a function of input power. In this planning-level model, η_E was treated as a nominal efficiency (LHV basis), consistent with typical PEM electrolyzer values reported in the literature, and implemented consistently through the electrolyzer performance settings used in HOMER.

In this study, a 10 kW electrolyzer was implemented for hydrogen generation purposes [25]. The detailed technical characteristics of the unit are outlined in Table 2.

2.1.3. Hydrogen tank

A hydrogen storage unit is designed to securely hold hydrogen gas for future energy needs. Within renewable energy frameworks, it serves as a buffer by retaining hydrogen produced from excess electricity, ensuring energy availability during periods of low generation. The effectiveness and reliability of such a system rely on its construction materials, pressure-handling capability, and the chosen storage form, whether gaseous, liquid, or solid. Pressurized containers are widely employed for gas-phase storage, while specialized cryogenic systems are necessary to maintain hydrogen in its liquid state due to its extremely low temperature requirements. This stored hydrogen can be reconverted into electricity via a fuel cell or used in various sectors such as industry and transportation, establishing hydrogen tanks as key assets in clean energy infrastructure. The total energy stored in the tank can be described mathematically by Eq. (3) [27].

$$HT_E(t) = HT_E(t-1) + \left(P_E(t) - \frac{P_{HT/FC}(t)}{\eta_{HT}}\right) \times \Delta t \quad (3)$$

where $HT_E(t)$ is the stored hydrogen energy (J), Δt is the simulation time step (s), η_{HT} is the storage efficiency, and the charging/discharging terms account for electrolyzer hydrogen input and fuel-cell hydrogen withdrawal. The tank state was constrained between empty and maximum storage capacity. $P_{HT/FC}$ is the hydrogen chemical power withdrawn from the tank and sent to the fuel cell, as defined in the subsection 2.1.1.

In the simulation process, a hydrogen storage unit with a 40-liter capacity was implemented [26]. The technical details and characteristics associated with this component are outlined in Table 3. The costs reported for the hydrogen tank in Table 3 (capital and replacement costs) correspond to the pressure vessel only. Since the storage pressure was set to 150 bar, hydrogen must be delivered at this pressure level. In practice, PEM electrolyzers commonly operate at hydrogen outlet pressures on the order of 30–40 bar, therefore an additional compression stage is typically required to store hydrogen at 150 bar unless a high-pressure electrolyzer is used. In this study, the compression requirement was accounted for by considering a compressor (or an equivalent integrated compression function) in the hydrogen storage chain, and its electricity demand was treated as an auxiliary consumption associated with hydrogen charging.

Table 2
Technical characteristics of the electrolyzer [25].

Parameters	Values
Model	JY-hydrogen generator
Noise Level	Low
Purpose	Gas Separation
Hydrogen Capacity	5–1000 nm^3/Hr
Hydrogen Purity	99%–99.99%
Specification	1200 mm × 850 mm × 1600 mm – 3200 mm × 1800 mm × 3500 mm
Capital cost	20,000 US$
Replacement cost	20,000 US$
Operation and maintenance cost	200 US$

Table 3
Technical characteristics of the hydrogen tank [26].

Parameters	Values
Material	37 Mn
Capacity	40 L
Working pressure	150 Bar
Compressor included in storage chain	Yes (externa/integrated)
Electrolyzer outlet pressure assumption	30–40 bar
Weight	47 ∓ 1 kgs
Specification	232 mm × 1180 mm
Wall thickness	6.0 mm
Pressure	10.0 MPa ≤ p < 100.0 MPa
Pressure level	High Pressure (10.0 MPa ≤ p < 100.0 MPa)
Test pressure	250 Bar
Outside diameter	232 mm
Capital cost	100 US$
Replacement cost	100 US$
Operation and maintenance cost	0 US$

2.1.4. Converter

In integrated energy systems, a converter functions as a key device that enables the transformation of electrical energy from one form to another, ensuring proper operation among all components. It generally consists of an inverter, which changes direct current from elements like batteries or renewable sources into alternating current suitable for loads, and a rectifier, which performs the opposite when necessary. The converter's performance is influenced by aspects such as load conditions, cooling mechanisms, and switching mechanisms. Within renewable-based configurations, converters contribute to maintaining voltage stability, enhancing energy transfer, and supporting seamless interaction between energy generation and storage units. The required converter capacity is determined by peak demand and conversion efficiency, as formulated in Eq. (4) [27].

$$P_{conv}(t) = \frac{P_{m,l}(t)}{\eta_{inv}} \quad (4)$$

where P_{conv} is the converter rated power (W), $P_{m,l}$ is the peak electrical demand (W), and η_{inv} is the inverter efficiency. The converter capacity was selected to satisfy peak demand while accounting for conversion losses.

A converter rated at 30 kW was selected for this analysis [28]. Its technical characteristics are outlined in Table 4.

2.1.5. Steam turbine generator

In this study, the steam turbine generator was assumed to be supplied

Table 4
Technical characteristics of the converter [28].

Parameters	Values
Nominal power	30 kW
DC side A voltage range	0Vdc – 60Vdc
DC side A max current	500Adc
DC side B voltage range	up to 900Vdc
DC side B max current	32Adc
Galvanic isolation	Yes
Efficiency	95%
Energy transfer direction	Bidirectional
Cooling	Liquid
Mehanical dimensions (W × L × H)	300 mm × 300 mm × 110 mm
Weight	18.1 kg
Capital cost	18,000 US$
Replacement cost	18,000 US$
Operation and maintenance cost	0 US$

by a geothermal steam cycle at Mount Cameroon. Hot geothermal fluid was produced to the surface through a production well. Because the deep fluid is maintained in a liquid state by high pressure, a controlled pressure drop at the surface caused partial flashing to steam. A flash separator then separated the steam phase from the remaining brine. The separated steam was routed to the steam turbine, expanded through the turbine stages to produce mechanical work, and converted to electricity by the coupled generator. After expansion, the exhaust steam was condensed, and both the condensate and the separated brine were reinjected into the reservoir to sustain the resource and minimize environmental discharge. In line with the planning-level scope of this work, detailed reservoir/wellbore calculations were not modeled explicitly; instead, the geothermal plant was represented in HOMER Pro as a steady generator, while the above description provides the physical basis for the steam supply.

A steam turbine generator is a device that converts thermal energy from pressurized steam into mechanical energy and subsequently into electrical energy through an integrated generator. It operates based on the thermodynamic Rankine cycle, where high-pressure steam expands through turbine blades, causing them to rotate and drive the generator. The performance of a steam turbine generator depends on steam pressure, temperature, flow rate, and turbine efficiency. In hybrid energy systems, it can be powered by geothermal steam sources, making it a valuable component for sustainable power generation. The electrical output power of a steam turbine generator can be expressed in equation (5) [29]:

$$P_{out} = \dot{m} \times (h_{in} - h_{out}) \times \eta_t \tag{5}$$

where P_{out} is the electrical power output (W), $\dot{m}$ is the steam mass flow rate (kg/s), h_{in} and h_{out} are the inlet/outlet steam specific enthalpies (kJ/kg), and η_t is the turbine efficiency. The generator output was constrained by its rated power.

In Eq. (5), the inlet/outlet steam enthalpies and steam mass flow rate are determined by the assumed geothermal steam supply conditions (flash/separator pressure and steam quality), while HOMER Pro represents the resulting geothermal plant as a generator with the selected rated power and operational constraints.

A steam turbine generator rated at 110 kW was selected for this analysis [30]. Its technical characteristics are outlined in Table 5.

2.1.6. Economic modeling

The economic assessment and optimization were performed in HOMER Pro using lifecycle cost accounting over a project lifetime R_{proj}. HOMER computes the total net present cost $C_{\text{NPC,tot}}$ as the present value of all capital, replacement, operation and maintenance, fuel (if any), and other costs over the project lifetime, minus revenues and the salvage value at the end of the project. HOMER then converts lifecycle costs into an equivalent total annualized cost $C_{\text{an,t}}$ using the capital recovery factor $CRF(i, R_{\text{proj}})$, where i is the annual real discount rate derived from the nominal discount rate and expected inflation. It can be defined in Eq. (6) [27,43–47]:

$$C_{\text{an,t}} = C_{\text{NPC,tot}} \bullet CRF(i, R_{\text{proj}})$$

with

$$CRF(i, R_{\text{proj}}) = \frac{i(1+i)^{R_{\text{proj}}}}{(1+i)^{R_{\text{proj}}} - 1}$$

The levelized cost of energy (LCOE) was computed by dividing the annualized cost of producing electricity by the annual electrical load served. If no thermal load is considered, this reduces to Eq. (7) [27,43–47]:

$$LCOE = \frac{C_{\text{an, t}}}{E_s} \tag{7}$$

where E_s is the annual electrical energy served to the load. The optimization objective was to identify the system configuration that minimized $C_{\text{NPC,tot}}$ (equivalently $C_{\text{an, t}}$) while satisfying technical constraints (component ratings, operating limits, and energy-balance feasibility).

Hydrogen compression to 150 bar was included as an auxiliary subsystem; its CAPEX/replacement cost and electricity consumption were incorporated in the hydrogen storage chain cost/energy balance.

2.2. Case study

This section presents a detailed examination of the selected study area and the methodology applied to evaluate its suitability for geothermal-based hybrid energy deployment. It includes a geographical and socio-economic overview of the Mount Cameroon region, an analysis of geothermal resource availability using seismic imaging, as well as the environmental and social impacts of the proposed system. Additionally, a sensitivity analysis is conducted to assess how variations in key economic parameters influence system performance.

2.2.1. Study area description

Mount Cameroon is the tallest peak in Central Africa (4040 m above sea level) and is characterized by warm air circulating over its massive edifice. The Mount Cameroon area is influenced by warm, humid maritime air from the Gulf of Guinea. At the foothills/coastal belt (Limbe-Buea area), the near-surface air temperature typically lies in the mid-20 °C range across the year (approximately 23–31 °C depending on season and location). Because the site spans large elevation gradients, air temperature decreases with altitude; a representative lapse rate reported for Cameroon is about 5.4 °C per km, implying cooler conditions at higher elevations compared with the coastal plain. It is the only volcanic peak in Central Africa with ongoing recent eruptions (Fig. 1). Favalli et al. identified Mount Cameroon as one of Africa's most active effusive volcanoes [31]. Approximately 500,000 individuals living or working on its fertile slopes face significant risks from lava flow inundation.

Fig. 2 highlights the prominent elevation profile of Mount Cameroon, one of Africa's largest and most active volcanoes. Rising steeply from near sea level to over 4000 m, the volcano dominates the southwestern region of Cameroon. The high-elevation zone, shown in brown hues, indicates the summit area, while the surrounding green to blue gradients reflect the rapid change in elevation and the dense vegetation coverage at lower altitudes.

The hourly load profiles used in HOMER Pro were derived from a field survey conducted in 2024 in the target community. The survey collected (i) household/service-point characteristics, (ii) an inventory of electrical appliances and rated powers, and (iii) usage schedules (time-of-use) distinguishing daytime and evening peaks as well as weekday/weekend variations. A total of 120 respondents (households and/or service facilities) were interviewed using a structured questionnaire complemented by short observational checks. The raw survey data were converted into an hourly demand series using a bottom-up method, i.e., summing appliance power multiplied by reported operating hours for each time interval, and then aggregating across respondents to obtain representative daily profiles for each month. These monthly daily

Table 5
Technical characteristics of the steam turbine generator [30].

Parameters	Values
Manufacturer	Howden
Power output range	75–1000 kW
Weight	4500 kg
Space requirement	1.5 m × 2.5 m × 2 m (width × length × height)
Capital cost	2800 US$/kW
Replacement cost	2800 US$/kW
Operation and maintenance cost	1.7 US$/hour

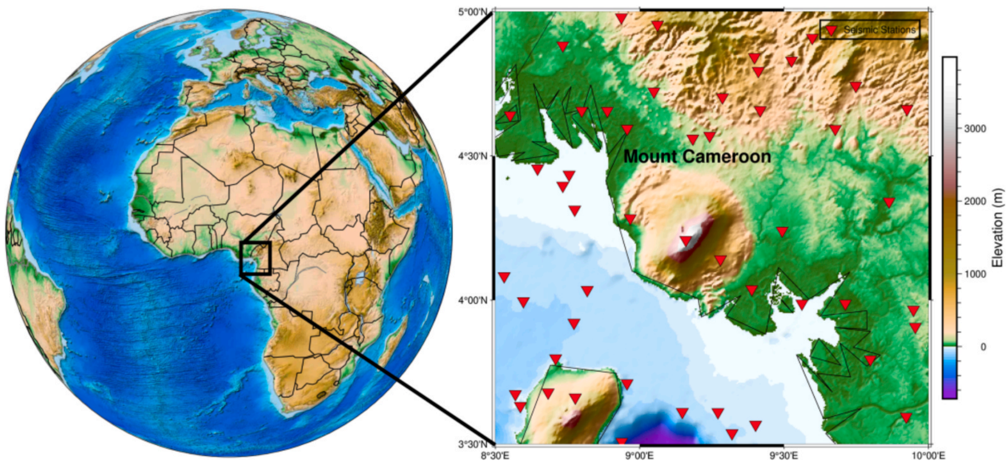

Fig. 1. Map of Africa Continent showing the location Mount Cameroon.

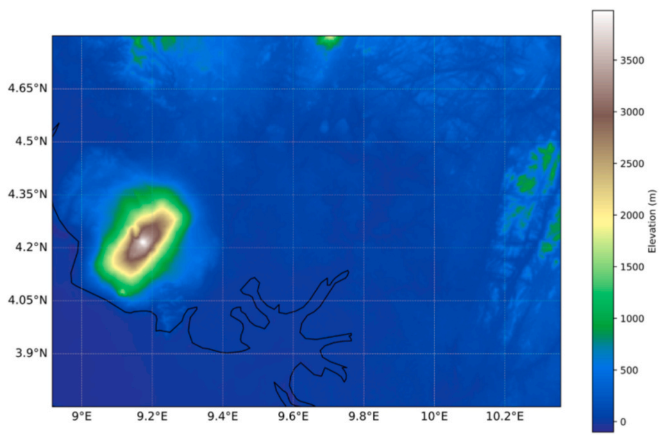

Fig. 2. Topography of Mount Cameroon.

profiles were used as the primary electrical-load input to HOMER (Fig. 3).

Fig. 3 illustrates the hourly variation in electricity demand across all twelve months, reflecting realistic energy consumption patterns of communities surrounding Mount Cameroon. A clear daily load fluctuation is observed, with three distinct demand peaks typically occurring

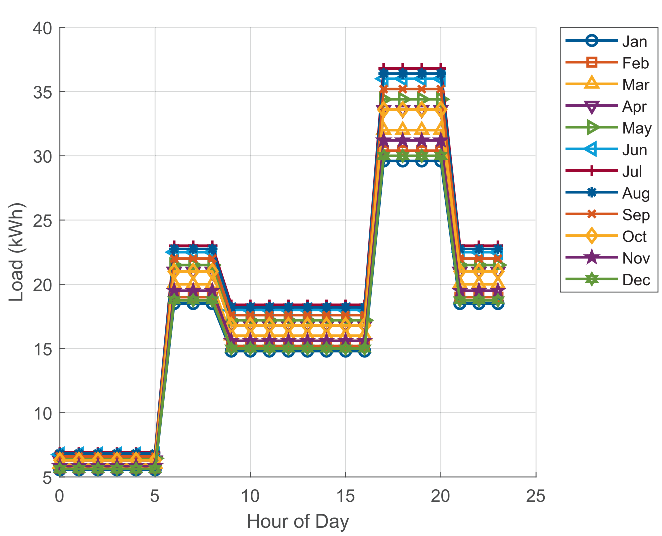

Fig. 3. Monthly daily load profiles derived from a 2024 survey.

in the early morning, midday, and evening hours. These variations indicate increased usage during key activity periods, likely corresponding to domestic lighting and appliance use, small business operations, food preservation, and healthcare services. The mid-day peak may also reflect energy needs in local schools, administrative offices, and markets. Seasonal differences, though subtle, suggest slightly higher energy requirements during the rainy months, potentially due to increased lighting and heating needs. This load profile realistically captures the dynamic and multifaceted energy needs of rural and semi-urban populations, emphasizing the importance of designing a flexible and resilient energy system capable of adapting to these temporal and seasonal consumption patterns.

Fig. 4 shows the architecture of a hybrid energy system designed to supply power and hydrogen to communities near Mount Cameroon. The system integrates a steam turbine generator, operating on geothermal heat to produce AC electricity, with an electrolyzer, which utilizes part of the generated power to produce hydrogen from water. The produced hydrogen is stored in a dedicated hydrogen tank for later use. When required, the stored hydrogen feeds a fuel cell, which generates DC electricity. A converter facilitates the bidirectional conversion between AC and DC to match the system's needs and ensure compatibility between components. The combined energy output, from the steam turbine and the fuel cell, is directed toward meeting the electricity demand of the surrounding population, including households, schools, health centers, and local businesses. This configuration provides a reliable, zero-carbon energy solution tailored to the region's volcanic geothermal potential and variable energy demand.

To make the operational logic explicit, Fig. 5 presents the hourly energy-management flowchart used to govern energy exchanges between the AC steam-turbine generator, the bidirectional converter, the DC electrolyzer/fuel-cell pair, and the hydrogen tank under SOC and rated-power constraints.

2.2.2. Geothermal potential assessment via seismic imaging

Seismic imaging is a geophysical technique that exploits the propagation of seismic waves to visualize the structure and properties of the Earth's subsurface [32]. Seismic waves are elastic waves generated by natural sources, such as earthquakes, or artificial sources, such as controlled explosions or mechanical vibrators. These waves travel through the Earth, interacting with different geological layers, where they are refracted, reflected, or scattered depending on the material properties encountered. The propagation of seismic waves is governed by the principles of elasticity and wave mechanics. There are two primary types of seismic waves: body waves and surface waves. Body waves, which travel through the Earth's interior, are further classified into P-waves (primary waves) and S-waves (secondary waves). P-waves

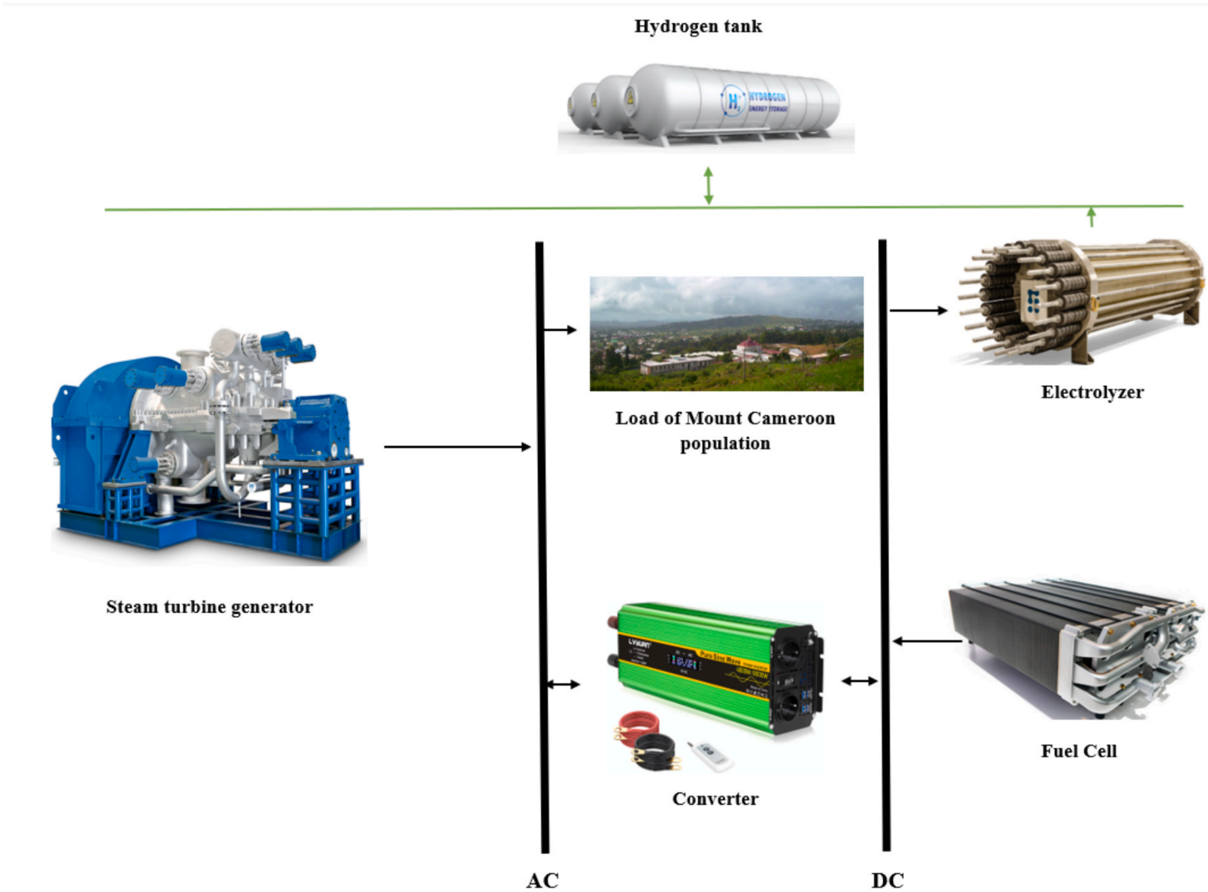

Fig. 4. Fuel Cell/Geothermal/Electrolyzer hybrid system architecture.

are compressional waves that move particles in the direction of wave propagation and are the fastest seismic waves, capable of traveling through solids, liquids, and gases. S-waves are shear waves that move particles perpendicular to the direction of propagation and can only travel through solids.

Surface waves, including Rayleigh waves and Love waves, propagate along the Earth's surface and typically have lower frequencies and higher amplitudes than body waves. While surface waves are valuable for imaging near-surface structures, body waves are more commonly used in seismic imaging for deeper subsurface exploration. Seismic imaging techniques leverage the travel time, amplitude, and frequency of seismic waves to infer the properties of subsurface materials. Variations in wave velocity and attenuation are indicative of changes in density, porosity, fluid content, and mineral composition of the underlying layers. These characteristics make seismic imaging an indispensable tool in various applications, including resource exploration, environmental studies, and hazard assessment.

2.2.3. *Environmental and social assessment*

The environmental and social assessment aims to evaluate the broader impacts of implementing the proposed Fuel Cell/Geothermal/Electrolyzer hybrid system in the Mount Cameroon region. From an environmental perspective, the system significantly reduces greenhouse gas emissions by utilizing renewable geothermal energy and green hydrogen, contributing to climate change mitigation and preserving local ecosystems. Socially, the project has the potential to enhance energy access for surrounding communities, particularly in remote areas with limited grid connectivity. It supports the development of essential services such as education, healthcare, and local businesses by providing a stable and clean energy supply. Additionally, the deployment of this system fosters job creation and skills development in the fields of renewable energy and hydrogen technology, promoting long-term socio-economic benefits for the local population.

2.2.4. *Sensitivity considerations*

The sensitivity considerations investigate the impact of varying key economic parameters on the overall performance and viability of the hybrid energy system. Specifically, different capital cost and replacement cost multipliers were applied to the fuel cell to assess how fluctuations in investment costs influence the system's net present cost and levelized cost of energy. The values of the capital cost and replacement cost multipliers of the steam turbine generator are given in Table 6.

3. Results and discussions

This section presents and analyzes the key findings of the study. It begins with a seismic imaging interpretation of Mount Cameroon, highlighting geothermal indicators critical to system design. The results then detail the optimal configuration of the proposed hybrid system based on technical and economic simulations. Subsequent subsections explore electricity and hydrogen production profiles, followed by an evaluation of the system’s environmental benefits and social relevance. Finally, sensitivity considerations assess the system’s economic resilience under varying cost assumptions, offering insights into its scalability and adaptability.

3.1. *Seismic imaging of Mount Cameroon*

Seismic low-velocity anomaly is well known to be source of long-term heat reservoir in the Earth [33]. Such a shear wave low-velocity imaged in East-Africa (below Afar and west of the Main Ethiopian Rift) has been investigated to have excess temperatures in the range of

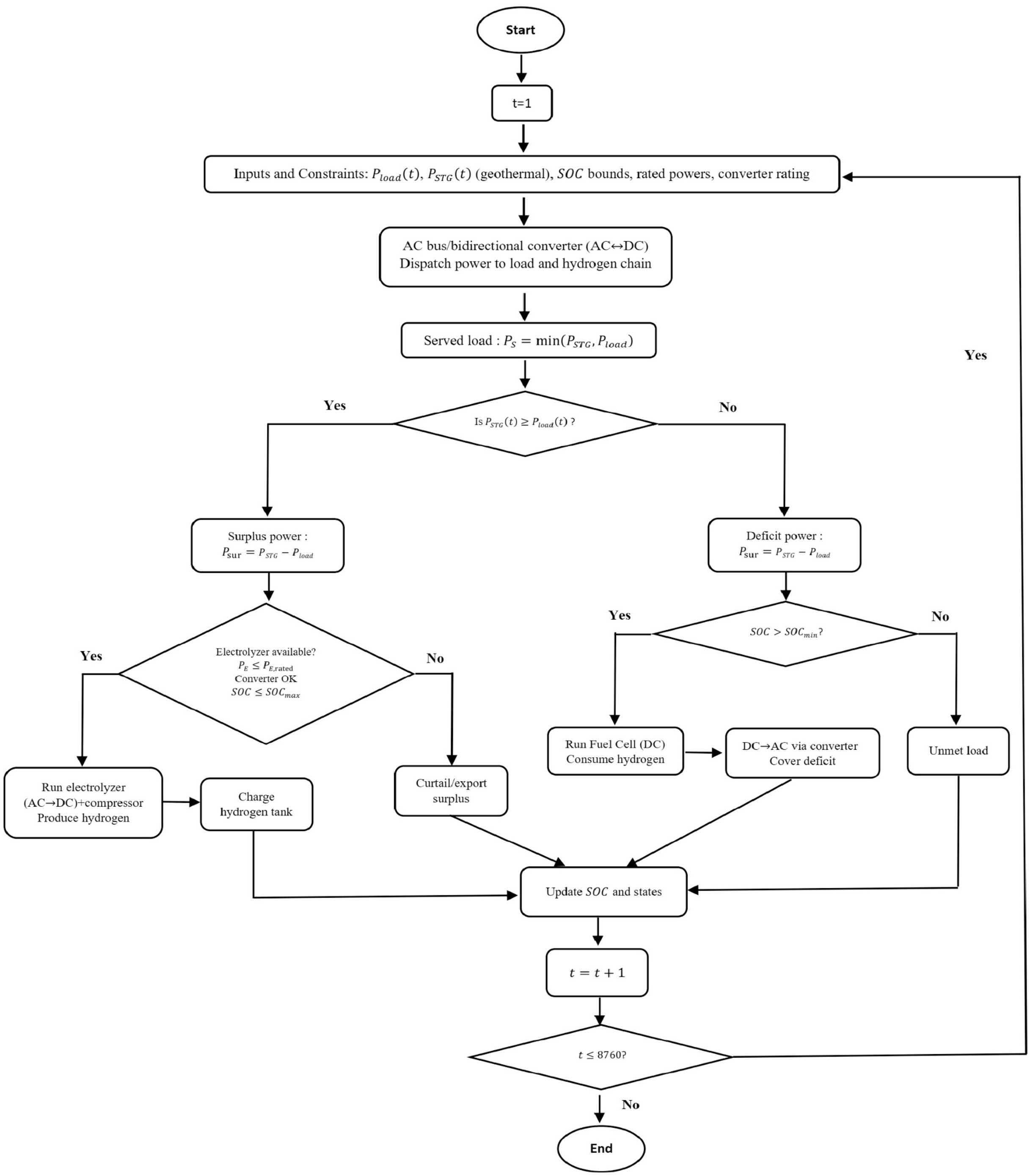

Fig. 5. Hourly energy-management flowchart of the geothermal-hydrogen hybrid system.

Table 6
Capital and replacement cost multipliers of the steam turbine generator.

Capital cost multiplier	Replacement cost multiplier
{0.50, 0.75, 1.00, 1.25, 1.50}	{0.50, 0.75, 1.00, 1.25, 1.50}

100–200 °C in the upper mantle [33]. The most impactful global tomographic models, SEMUCB-WM1 [35] and GLADM25 [34] obtained through different full-waveform inversion approach reveal a significant low shear wave velocity anomaly beneath Mount Cameroon, located at the base of the Earth's crust (Fig. 6). This seismic shear wave low-velocity anomaly beneath volcanic hotspot is typically associated with regions of high temperature and partial melt [36,37], which are key indicators of deep mantle upwellings and magmatic processes. The presence of such a low-velocity zone suggests the existence of a long-term thermal reservoir that could be actively supplying heat and possibly influencing the volcanic and geothermal activity of the region. These anomalies are often linked to deep mantle plumes originating from upper mantle, mantle transition or core-mantle boundary [35,38], which is the heat transport from the Earth's interior to the surface. However, global seismic tomography models suffer from low resolution in this region, likely due to the limited density of seismic ray-paths crossing Mount Cameroon. This limitation can obscure finer-scale structures and thermal anomalies, making it challenging to accurately characterize the geothermal potential. To overcome these constraints,

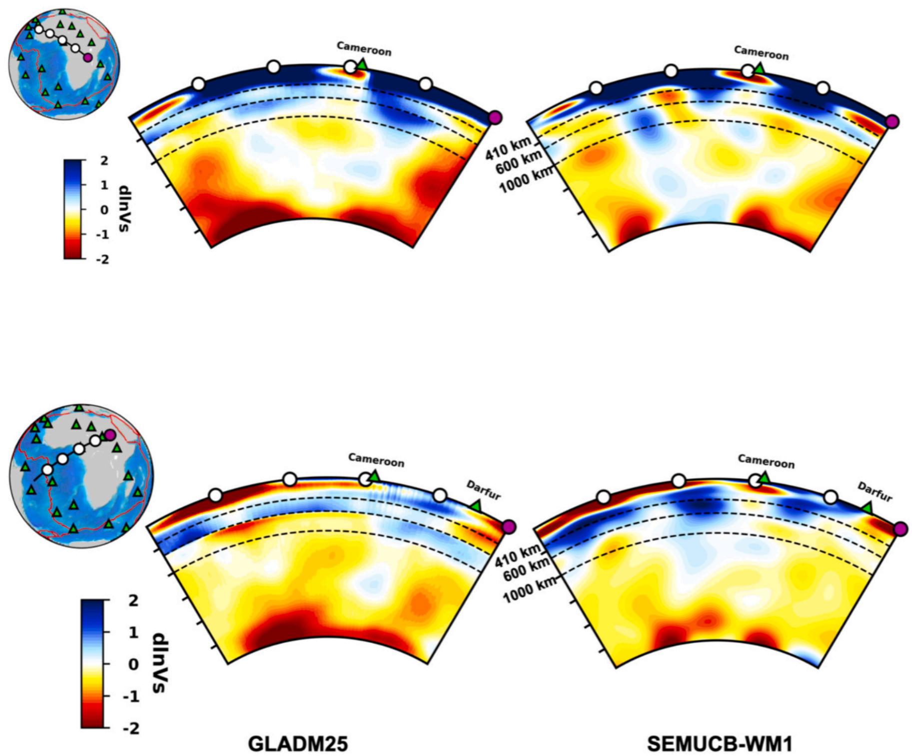

Fig. 6. Comparison of cross-sections across Mount Cameroon from two global tomographic models: GLADM25 [34] on the left, and SEMUCB-WM1 [35] on the right. These cross-sections extend from the surface down to the core-mantle boundary (CMB). The inset map shows the geographical line along which the cross-section was made. Triangles on the inset map represent hotspots. The dashed black lines in the cross-sections correspond to depths of 400, 660, and 1000 km.

regional high-resolution seismic imaging is essential, as it allows for a more detailed investigation of subsurface heterogeneities, including variations in temperature, composition, and melt distribution. A refined seismic tomographic model around Mount Cameroon would provide crucial insights into the thermal dynamics and lithospheric structure, offering a clearer understanding of its role in geothermal energy potential and broader geodynamic processes. Such high-resolution imaging could help determine whether the observed low-velocity anomalies correspond to localized magmatic intrusions, partial melting, or deep hydrothermal circulation systems. Such a study will enhance our ability to assess the geothermal potential of the region, which could have significant implications for renewable energy exploration and volcanic hazard assessment.

3.2. Optimal configuration of the hybrid system

The simulation results identified an optimized configuration that balances technical feasibility and economic performance for the Mount Cameroon context. The system comprises a 50 kW fuel cell, a 110 kW steam turbine generator, a 10 kW electrolyzer, a 25.5 kW converter, and a 20 kg hydrogen storage tank. This setup was selected for its ability to meet the load demand reliably while minimizing investment and operational costs. The net present cost of the system was estimated at US$ 1.84 million, with a levelized cost of energy of US$ 0.465/kWh. These results highlight the economic viability of integrating geothermal and hydrogen technologies for decentralized, clean energy supply in volcanic regions like Mount Cameroon.

3.3. Electricity production

Fig. 7 presents the monthly electricity production of the optimized hybrid system, highlighting the contributions of both the steam turbine generator (Gen) and the fuel cell (FC). The results indicate a relatively stable generation pattern throughout the year, with slight variations corresponding to seasonal resource availability and system efficiency. The fuel cell consistently provides a steady baseline energy supply, while the generator contributes additional power, particularly in months with higher energy demand. The highest production levels are observed between May and October, which suggests a seasonal influence on geothermal resource efficiency or hydrogen availability. These findings

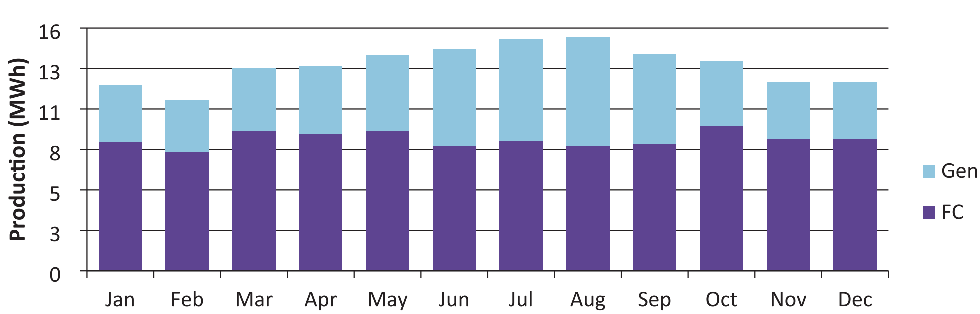

Fig. 7. Monthly electric production of fuel cell and steam turbine generator.

confirm the system's ability to meet energy demand reliably while maintaining operational flexibility. The hybrid integration of geothermal energy and hydrogen storage ensures continuous electricity supply, making this configuration suitable for decentralized energy solutions in volcanic regions like Mount Cameroon.

Fig. 8 shows the power output profile of the fuel cell throughout the year, with variations in electricity production depicted as a function of the time of day and day of the year. The color scale on the right indicates power levels, with dark purple areas representing low or zero output and yellow areas indicating peak production. The fuel cell operates primarily during specific periods of the day, with noticeable gaps in power generation during nighttime hours, likely due to reduced hydrogen availability or lower demand. Throughout the year, fluctuations in power output reflect seasonal variations in energy consumption or system efficiency. The intermittency observed suggests that the fuel cell dynamically adjusts its operation based on load requirements and hydrogen storage levels, reinforcing its role as a flexible and responsive energy source within the hybrid system.

Fig. 9 highlights the power output profile of the steam turbine generator over the course of a year, with power variations displayed as a function of both the time of day and day of the year. The color scale on the right represents power levels, where dark purple indicates low or zero output, and yellow signifies peak generation. The steam turbine operates predominantly in the late afternoon and early evening, with sporadic activity during other hours of the day. The consistent band of higher output around 18:00 suggests that the turbine is primarily utilized during peak demand periods, likely to supplement fuel cells. Additionally, intermittent generation in the early morning hours indicates occasional activation, possibly influenced by seasonal variations in thermal energy availability or load requirements.

The converter's power output varies throughout the year, with distinct operational patterns observed at different times of the day (Fig. 10). Its activity is primarily concentrated between 6:00 and 18:00, aligning with peak energy demand periods. A noticeable drop or shutdown occurs around 18:00, indicating either reduced demand or a shift in energy distribution within the hybrid system. The variations in output, particularly around midday, suggest that the converter dynamically responds to fluctuations in energy generation from geothermal and fuel cell sources. The observed trends highlight the converter's role in managing power flows and ensuring compatibility between AC and DC components, ultimately contributing to system stability and efficiency.

3.4. Hydrogen production and consumption

The hydrogen production profile exhibits a gradual increase over the year, with a clear transition from low to high production levels around mid-year (Fig. 11). Initially, hydrogen production remains relatively low, fluctuating between 4 kg and 8 kg, indicating limited surplus energy available for electrolysis. As the year progresses, production steadily increases, reaching peak values of approximately 20 kg during the latter half of the year. This trend suggests that energy availability from the hybrid system improves over time, possibly due to seasonal variations in geothermal and fuel cell performance. The uniform distribution across all hours of the day highlights continuous hydrogen generation, ensuring a stable supply for storage and later use in the system.

Fig. 12 presents the annual time-of-use map of hydrogen consumption (hour of day vs. day of year), where the color scale indicates the instantaneous consumption rate (0 to ~1.6 L/h). Overall, hydrogen use was non-uniform over both the day and the year, reflecting the dispatch strategy of the system.

Two main patterns were evident:

- Strong diurnal structure: Hydrogen consumption was generally low during the early hours, then increased during the daytime and evening periods. This indicates that the fuel cell was called more frequently during hours when the electrical deficit was higher, i.e., when the steam turbine generator contribution were insufficient to fully meet demand.
- Persistent high-consumption band: A distinct high-consumption region appeared around the late-day to night window across most days of the year. This suggests recurrent operation near the upper range of hydrogen consumption during those hours, consistent with sustained deficit compensation.

From a sizing/feasibility standpoint, Fig. 12 supports the interpretation that hydrogen was repeatedly mobilized as a balancing resource throughout the year, with peak usage episodes reaching the upper bound of the observed range.

3.5. Environmental and social analysis

The environmental and social analysis provides crucial insights into the feasibility and acceptance of the proposed Fuel Cell/Geothermal/Electrolyzer hybrid system among communities living near Mount Cameroon. The survey results indicate a strong local interest in renewable energy solutions, with a significant majority of respondents recognizing the potential benefits of geothermal energy in reducing dependence on fossil fuels and minimizing environmental degradation. Socially, the system is perceived as a driver for economic growth, offering employment opportunities in system operation, and maintenance. Additionally, concerns regarding affordability and accessibility were highlighted, emphasizing the need for policy support and financing mechanisms to ensure inclusivity. Environmentally, respondents acknowledged the system’s capacity to deliver clean energy with zero greenhouse gas emissions, reducing deforestation and indoor air pollution associated with traditional biomass use.

3.6. Sensitivity considerations

The sensitivity considerations provide a comprehensive evaluation of the economic viability and resilience of the proposed Fuel Cell/Geothermal/Electrolyzer hybrid system under varying financial conditions. By considering multiple capital cost and replacement cost multipliers (0.50, 0.75, 1.00, 1.25, and 1.50), the study assesses the impact of investment fluctuations on the system’s performance. The results reveal that while a decrease in capital costs significantly improves the system’s

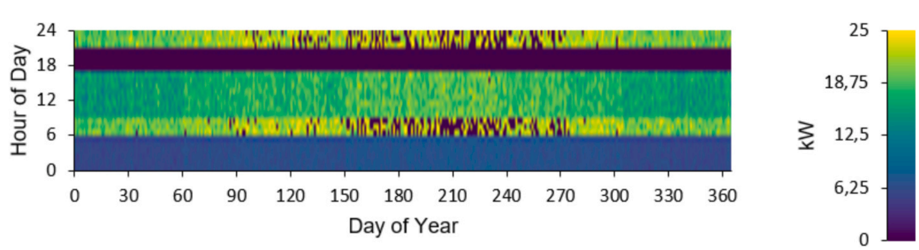

Fig. 8. Power output from the fuel cell.

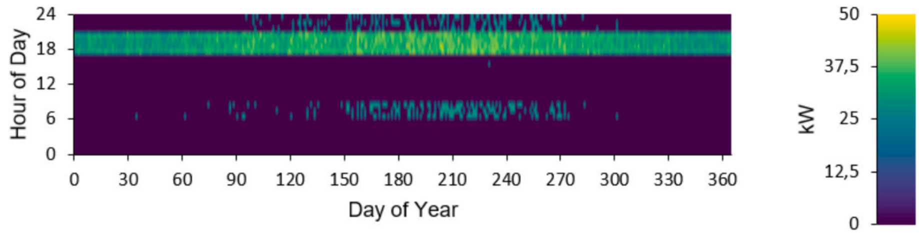

Fig. 9. Power output from the steam turbine generator.

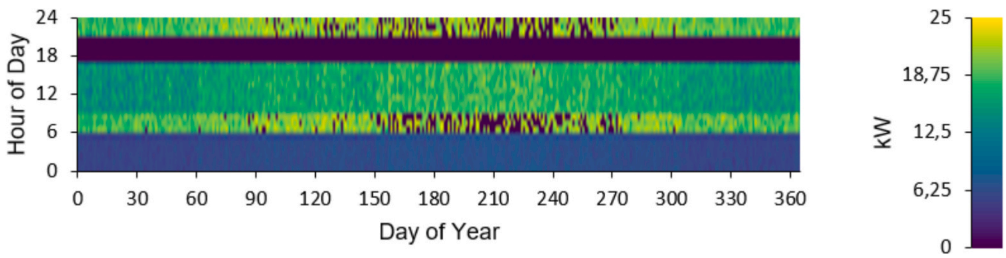

Fig. 10. Power output from the converter.

economic feasibility by lowering the net present cost and levelized cost of energy, an increase in these costs raises financial constraints, making the system less competitive. They also highlight the crucial role of hydrogen storage in stabilizing energy supply and optimizing costs over the system's lifetime. Furthermore, variations in fuel cell costs demonstrate that the hybrid system remains viable within a certain range of cost adjustments, reinforcing its adaptability to different market conditions.

4. Conclusion

This study provided a comprehensive technical, economic, environmental, and social evaluation of a fuel cell, geothermal, and electrolyzer hybrid system for hydrogen and electricity production in Mount Cameroon. Seismic imaging identified high-temperature geothermal reservoirs, confirming the feasibility of geothermal exploitation for sustainable energy generation. The optimal system configuration consisted of a 50 kW fuel cell, a 110 kW steam turbine generator, a 10 kW electrolyzer, a 25.5 kW converter, and a 20 kg hydrogen tank, achieving a net present cost of $1.84 million and a levelized cost of energy of $0.465/kWh. This cost was competitive compared to conventional off-grid fossil fuel systems and highlighted the economic feasibility of hydrogen-based energy solutions in volcanic regions. The system ensured stable and continuous power generation, with the fuel cell and steam turbine providing complementary electricity outputs to meet the varying demand of local communities, while the electrolyzer effectively utilized excess energy to produce and store hydrogen, enhancing grid flexibility and energy security. The environmental assessment confirmed the system's zero greenhouse gas emissions, offering a clean alternative to diesel generators, while the social analysis indicated strong community support due to employment opportunities, improved energy access, and local economic development. Sensitivity considerations revealed that reducing the capital cost of fuel cells and geothermal turbines significantly improved system affordability, with

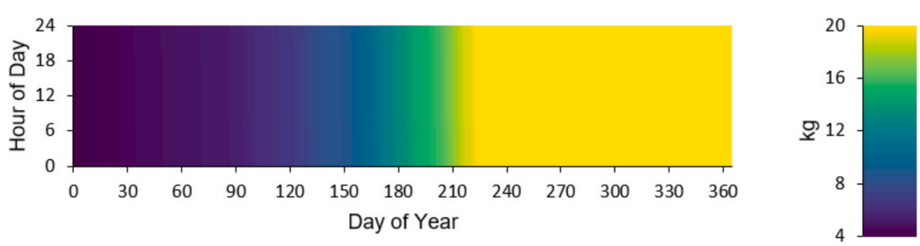

Fig. 11. Annual hydrogen production.

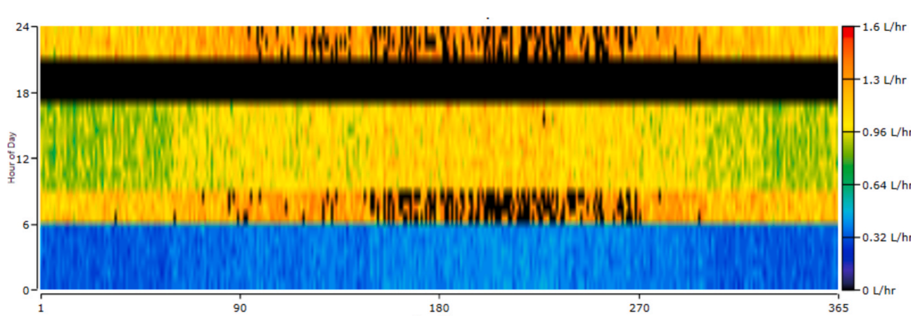

Fig. 12. Annual hydrogen consumption.

hydrogen storage playing a key role in cost stabilization and long-term energy supply resilience. Future research should explore advanced geothermal drilling techniques to optimize heat extraction efficiency, next-generation electrolyzers with higher conversion rates, and artificial intelligence-based energy management for improved real-time load balancing. Additionally, policy incentives and financial support mechanisms should be developed to lower the initial capital cost of hydrogen technologies, making them more accessible for rural electrification. A techno-economic comparison with other renewable energy sources, such as solar and wind, would provide deeper insights into the most cost-effective and sustainable solutions for off-grid communities.

CRediT authorship contribution statement

Pascalin Tiam Kapen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **M.D. Wamba:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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